

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MECHANICAL PROPERTIES OF FLUIDS

LECTURE-02 **PHYSICS**

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1** Viscosity, Stroke's Law
- 2** Terminal Velocity
- 3** Application of Pascal Law **II**
- 4** Buoyancy/Buoyant Force
- 5** Archimedes Principle

RECAP FROM 1ST LECTURE

Fluid

liquid

gas

Density ρ

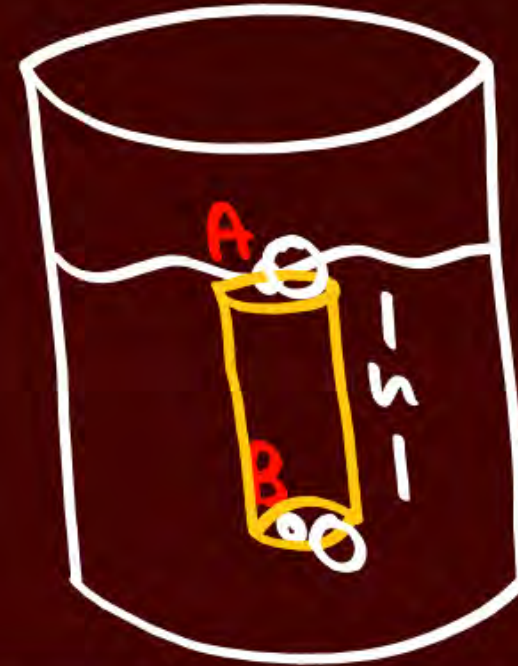
$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Atmospheric pressure P_0

Variation of pressure with depth



$$P_B = P_A + \rho gh$$

Pascal Law

Pascal law

Summer - 2026

(3M)

11

- A pressure applied to any point of a liquid, is transmitted equally to every portion of the liquid



✓ # $P_B = P_A + \rho gh$

$P_B = P_1 + P_A + \rho gh$

$P_B = P_1 + P_2 + P_A + \rho gh$

Application of Pascal law

12

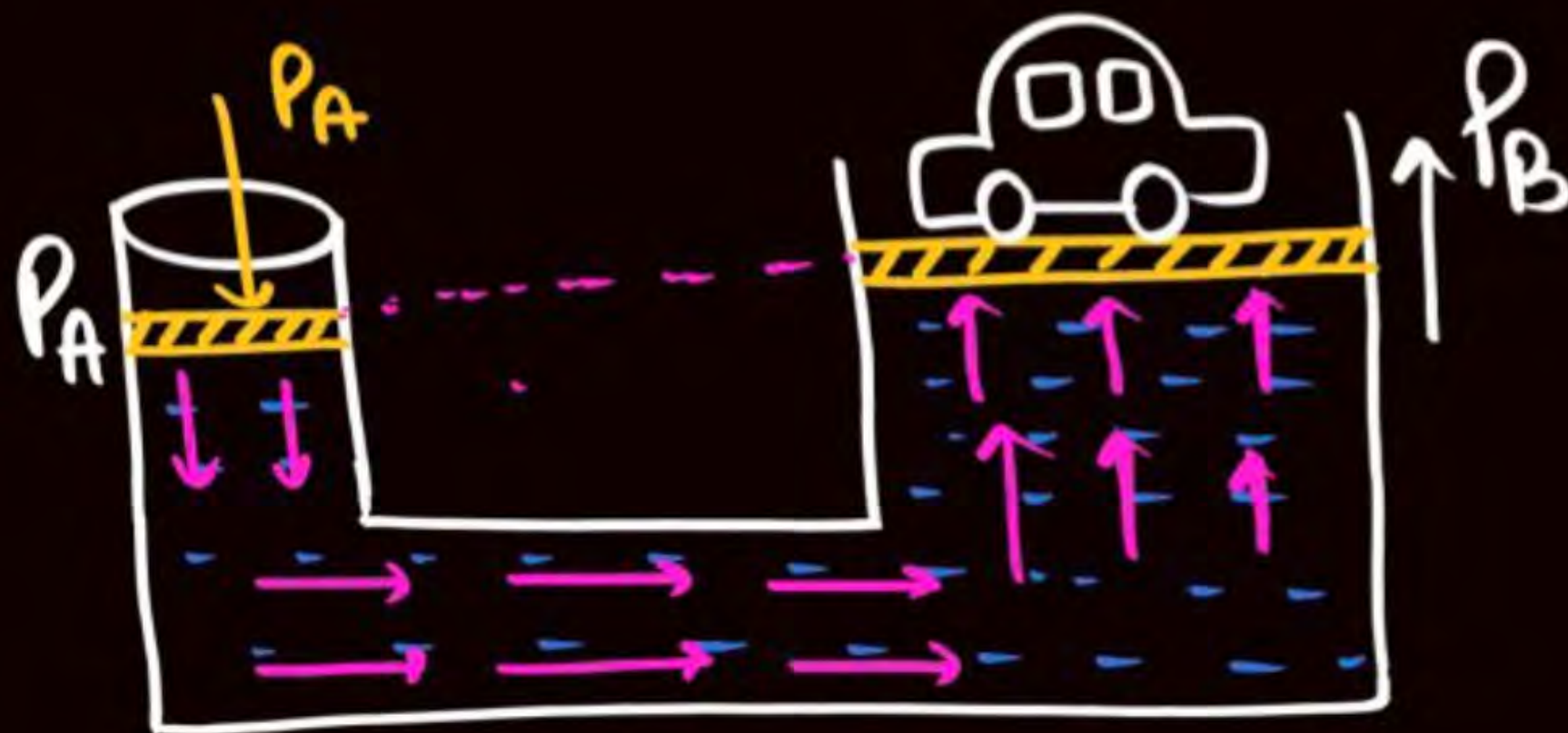
① Hydraulic lift

• Pascal law

qth

$$P_A = P_B$$

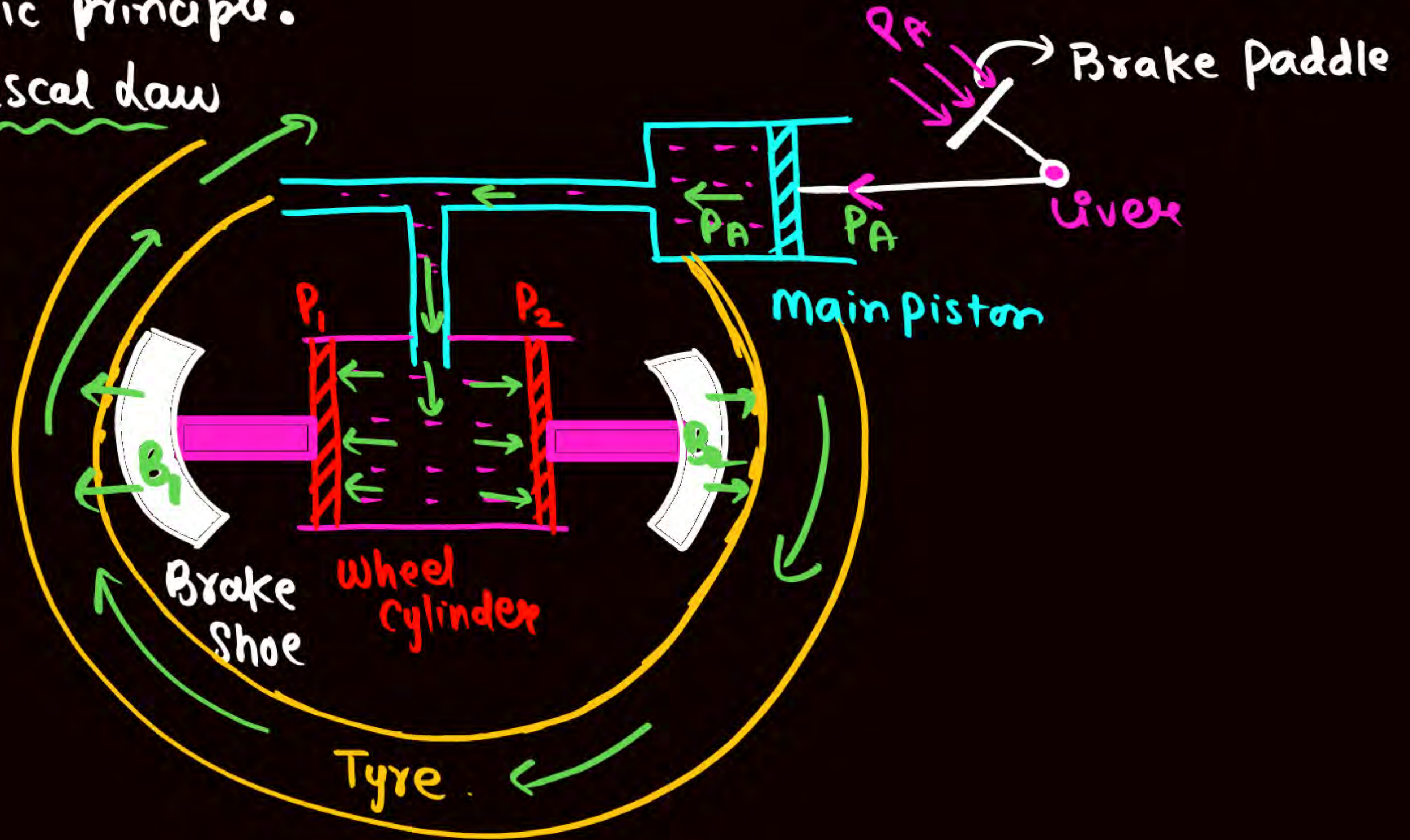
Piston



(2) Hydraulic Brakes.

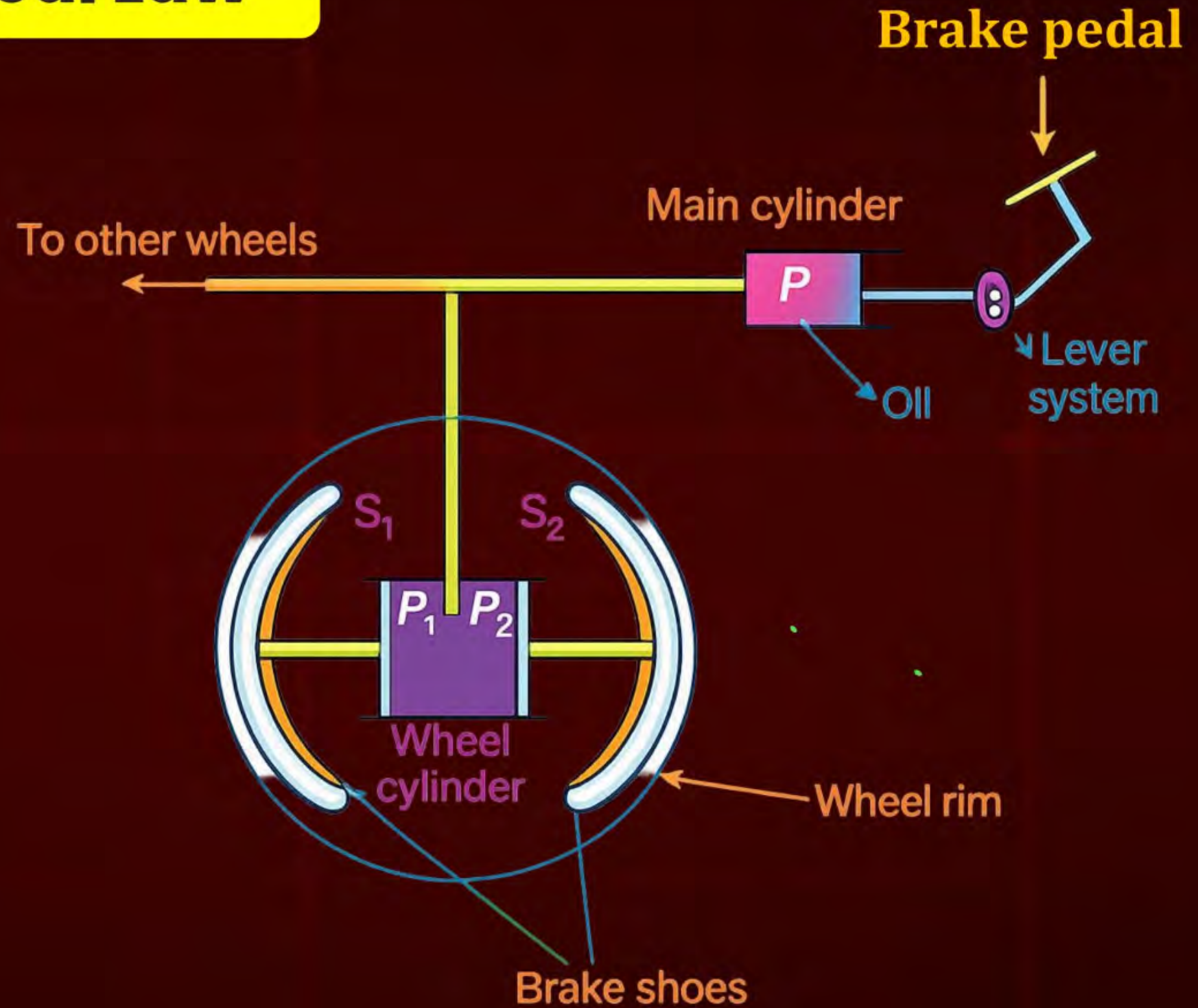
2.5 mark

Basic principle:
Pascal law



Application of Pascal Law

2. Hydraulic Brakes



Thrust (Perpendicular Force)

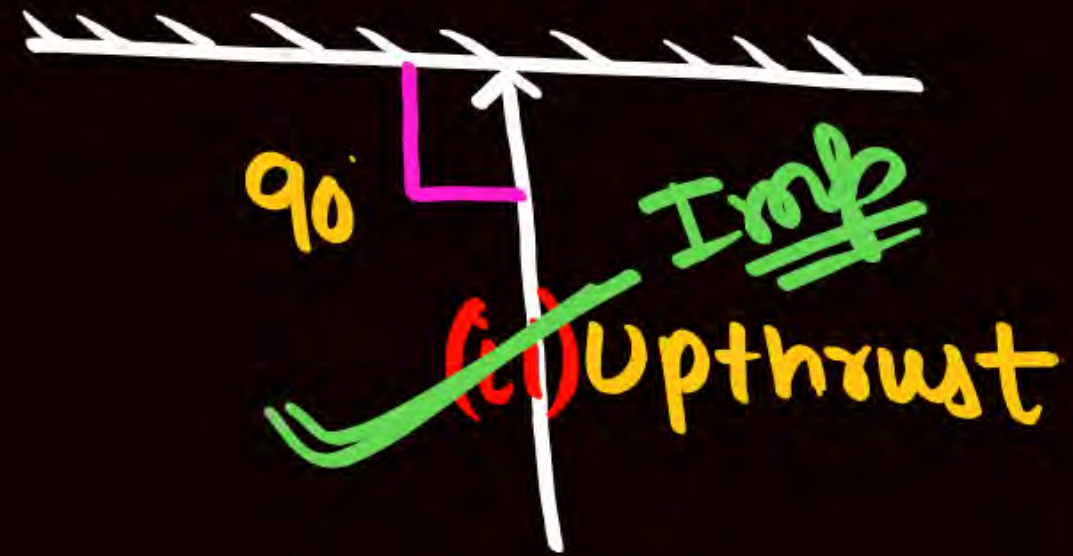
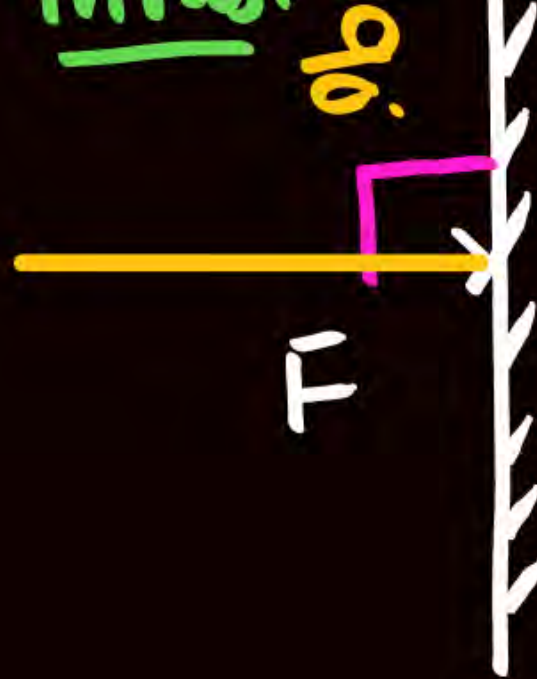
- Any perpendicular force is blan thrust

#

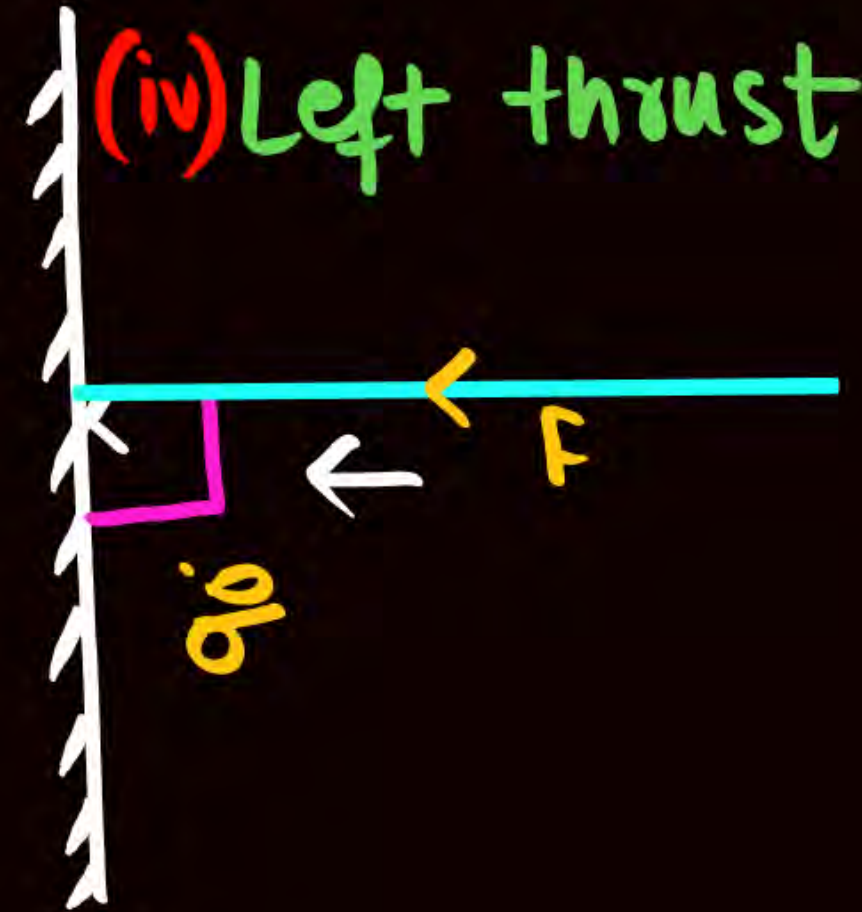
(i) down thrust



(iii) Right thrust



(iv) Left thrust

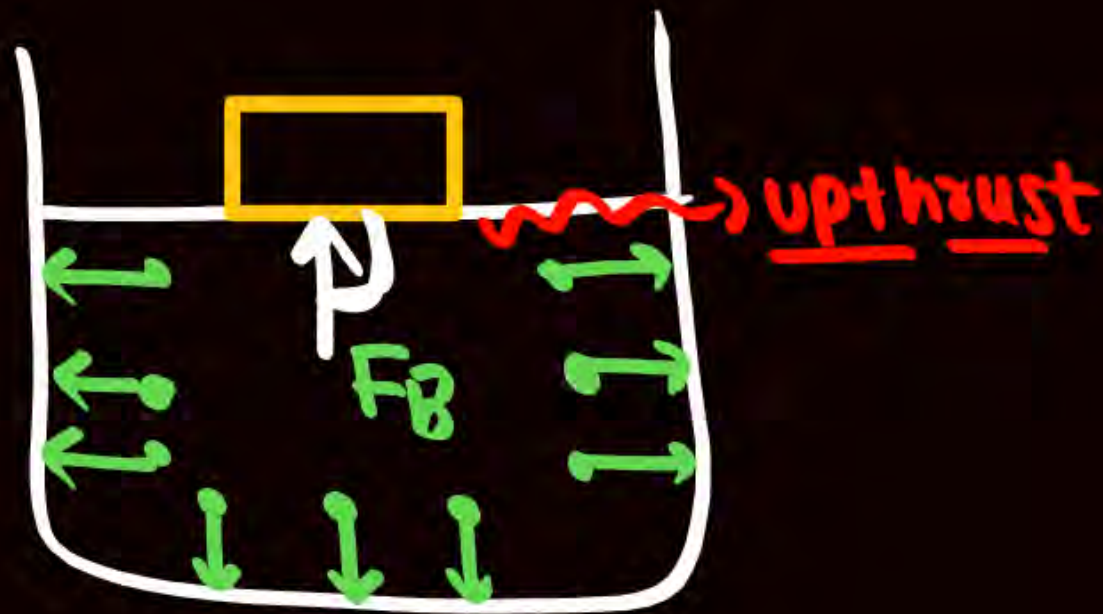


Buoyant Force

- the upthrust applied by a liquid on a object is called Buoyant force

• Buoyancy

It is the property of liquid by which it applies the Buoyant force

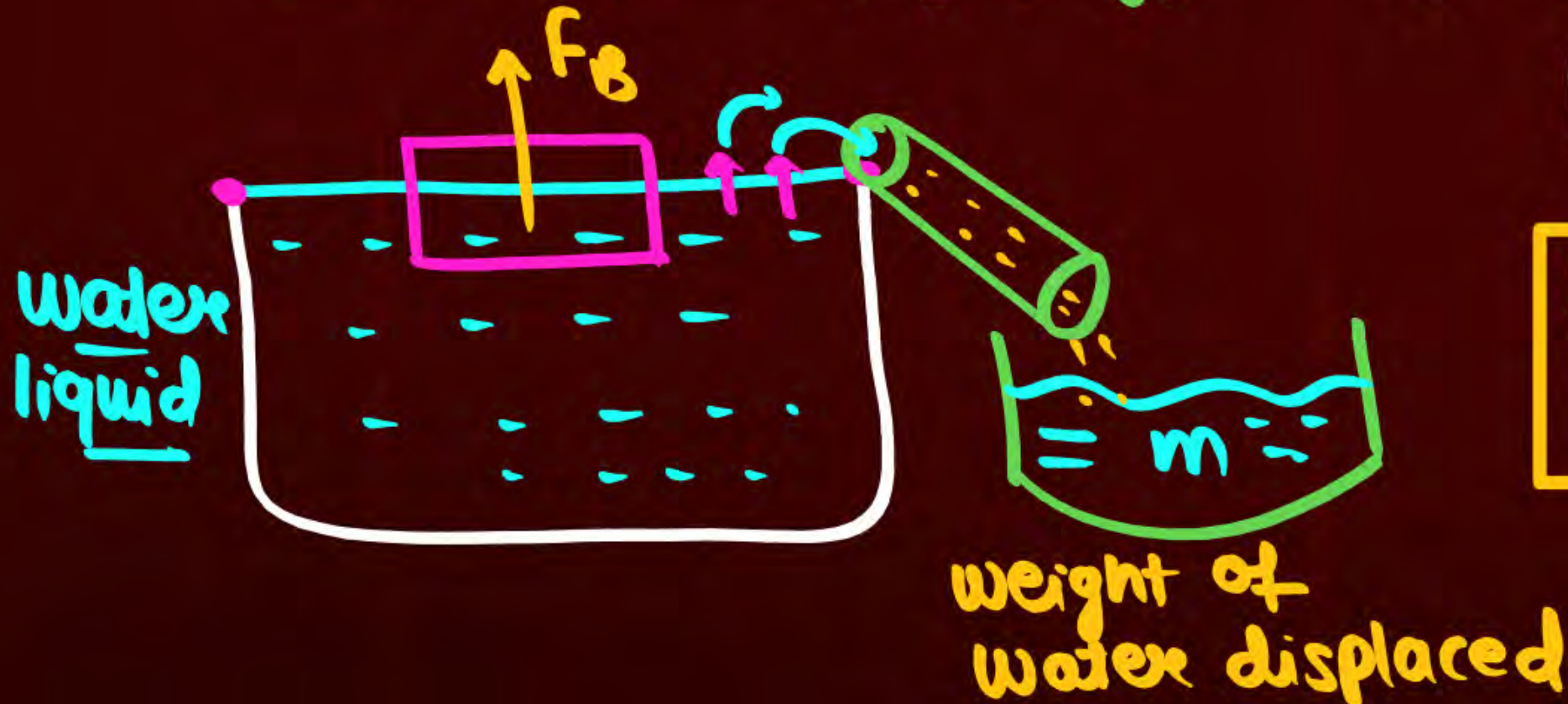


Archimedes Principle

- It give the magnitude of

Buoyant force

- It give the formula of Buoyant force ($F_B = \rho V g$)



" The buoyant force acting on an object is equal to the weight of water displaced.

$F_B = \text{weight of water (liquid) displace}$

$F_B = W$

$F_B = \underline{m}g$

$F_B = V \rho g$

$\rho = \frac{m}{V}$
 $V \rho = m$



Buoyant
Force

$$F_B = V \rho g$$

$V \Rightarrow$ volume of liquid displace.

$\rho \Rightarrow$ density of water

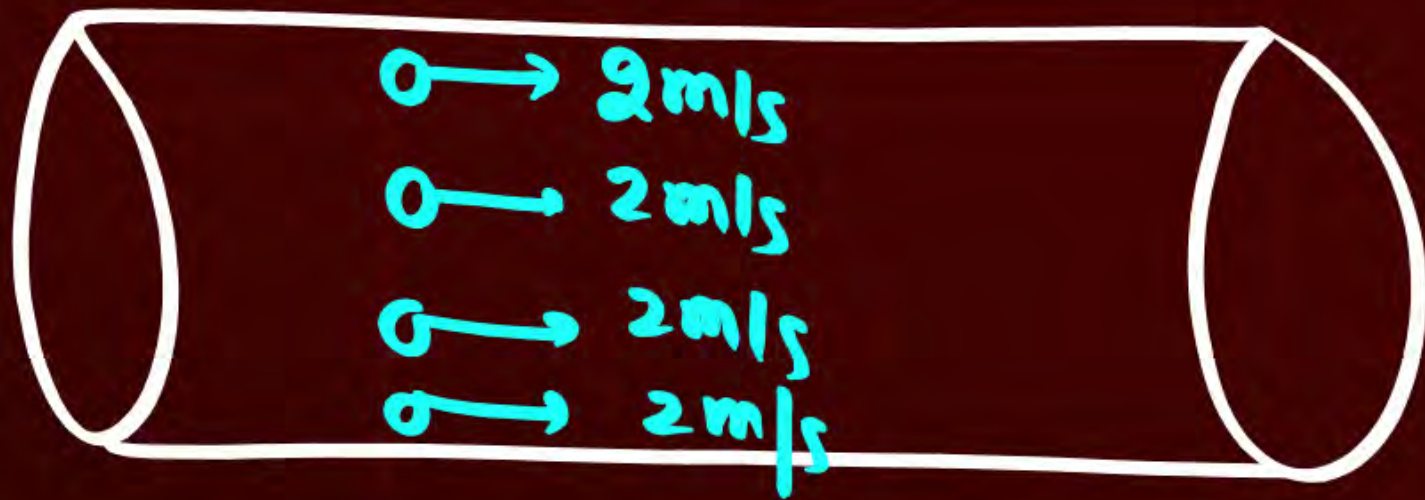
$g \Rightarrow$ acc. due to gravity

Streams And Turbulent Flow

Stream line flow

/ Laminar flow

- It is the type of flow in which particle follow the fixed path.



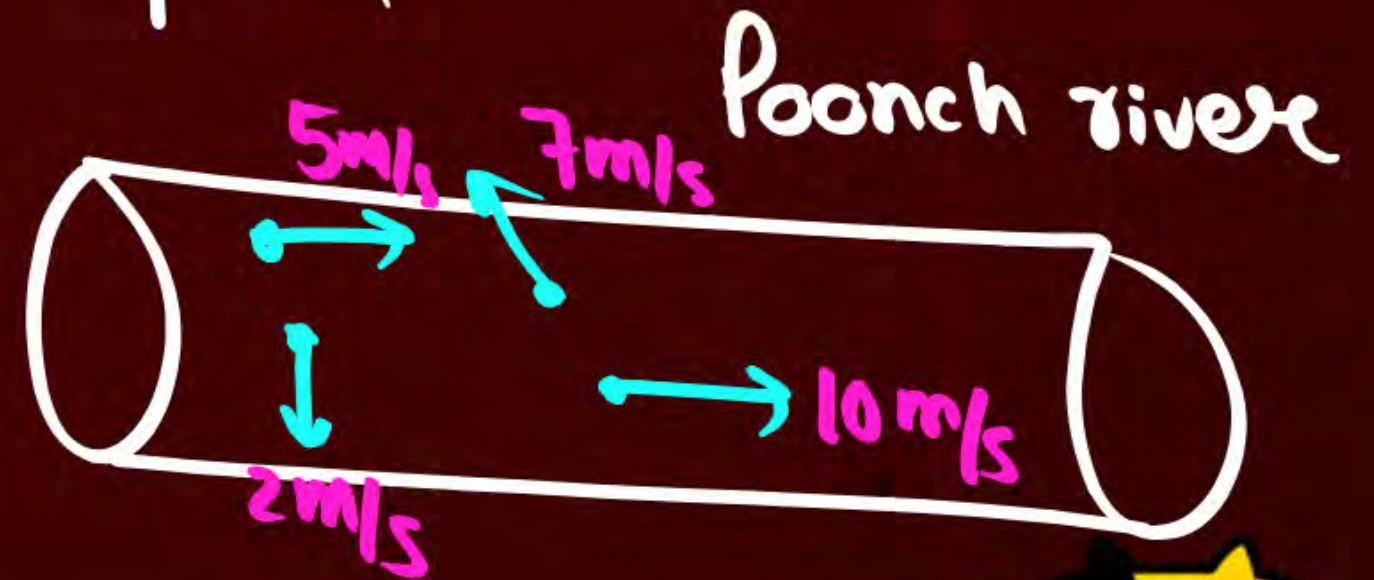
2024
2022

2021



Turbulent Flow

- It is the type of flow in which liquid does not follow the fixed



Reynold number

It tell us whether the
given flow is

Turbulent OR
Streamline flow

Neet-2021

$Re < 2000$ Stream line

Neet-2023

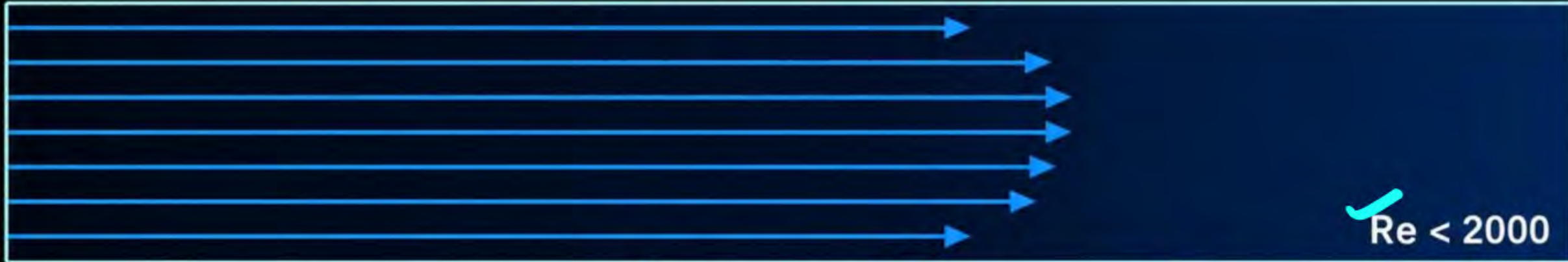
$Re > 2000$ Turbulent flow

Q. If Reynold number is 3500

- (a) flow 31.
- (b) Turbulent flow ✓
- (c) Streamline flow 81.
- (d) I donot known (I am donkey)
0.331.

Reynolds Number (Re)

Laminar Flow



Turbulent Flow



Re = 2000

$$Re = \frac{\rho V L}{\mu}$$

ρ : Density of the fluid

V : Velocity of flow

μ : Viscosity of the fluid

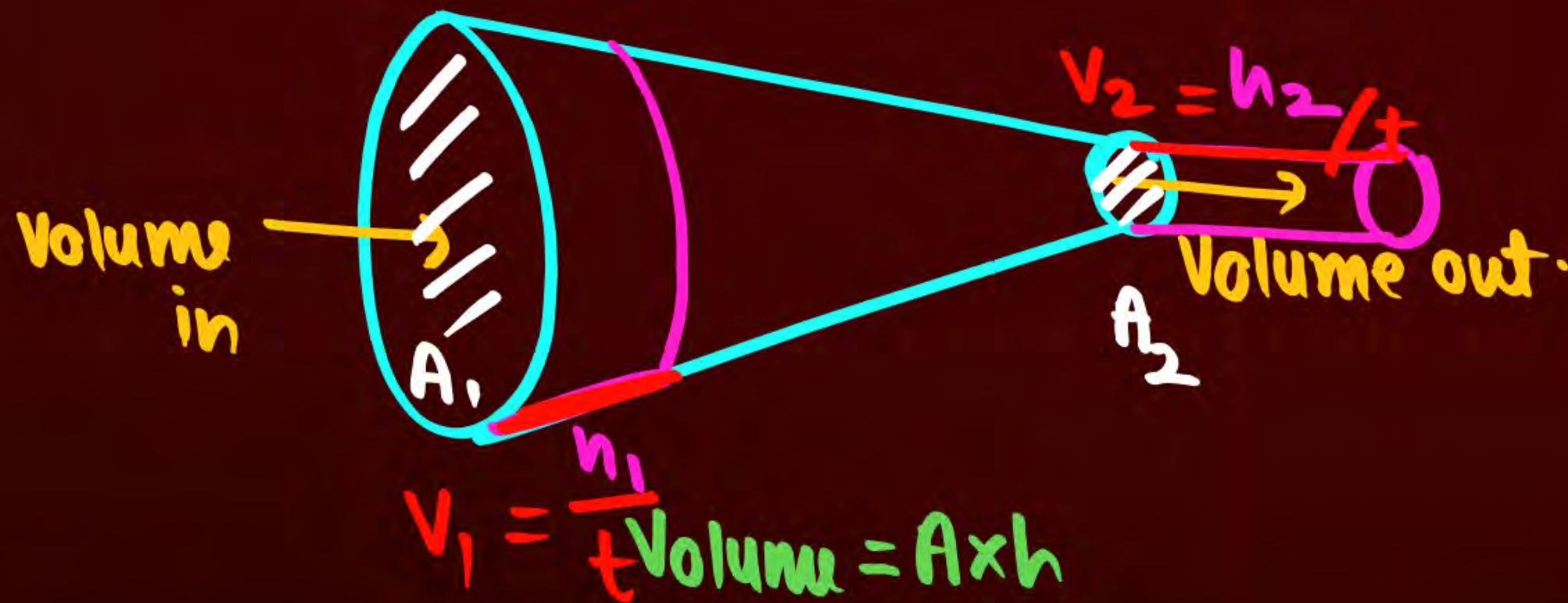
L : Characterizing length of the flow system
(usually the diameter of the pipe)

Equation of Continuity

"The volume of liquid in is equal to the volume of water out

$$A_1 > A_2$$

$$\begin{aligned} A &= l \times b \\ V &= l \times b \times h \\ \underline{V} &= \underline{A \times h} \\ &\quad \underline{12^{th}} \end{aligned}$$



Volume in = Volume out

$$A_1 \times h_1 = A_2 \times h_2$$

Divide b/s by (t)

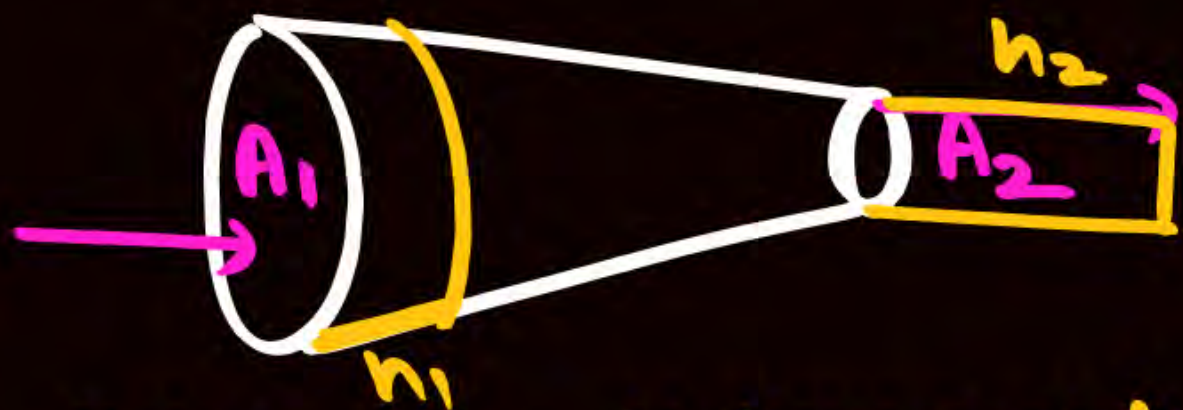
$$A_1 \times \frac{h_1}{t} = A_2 \times \frac{h_2}{t}$$

$$A_1 V_1 = A_2 V_2$$

Equation of continuity



Repeat



Volume in = Volume out

$$A_1 \times h_1 = A_2 \times h_2$$

$$A_1 \times \left(\frac{h_1}{t} \right)^{V_1} = A_2 \times \left(\frac{h_2}{t} \right)^{V_2}$$

$$A_1 \times V_1 = A_2 \times V_2$$

$$A_1 V_1 = A_2 V_2$$

Velocity

$$V_1 = \frac{A_2 V_2}{A_1}$$

Area

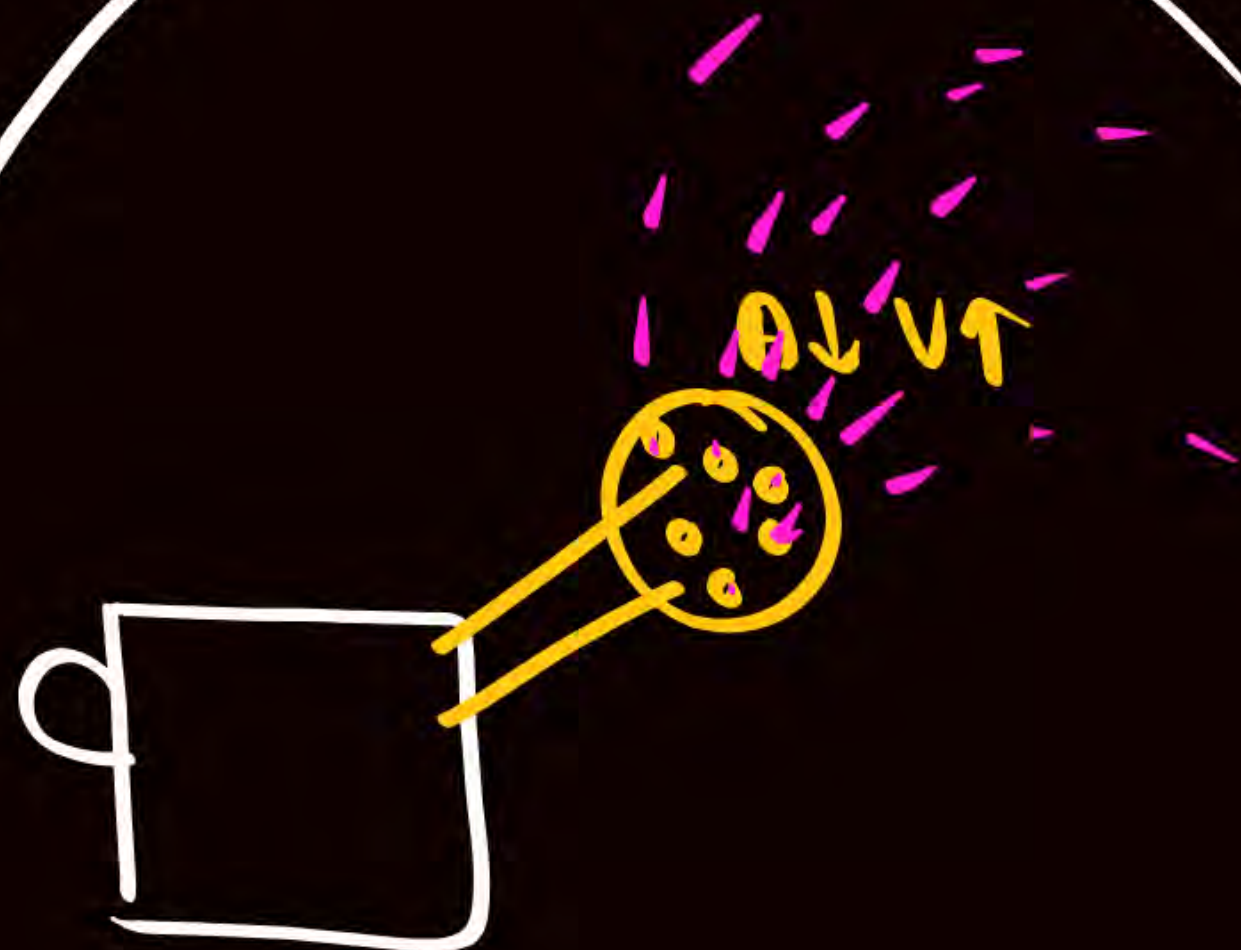
Velocity and Area is inversely Proportional to each other

$$\uparrow \downarrow A \propto \frac{1}{\downarrow \uparrow \uparrow V}$$

#



#



Lecture:-

Acche
emaandari

1 time
x2

Notes

Recall

1 chapter

Basic
Bohat
Acchi

NCERT Numerical X

HOMEWORK



Equation of Continuity ?

B- Theorem (2027)

SUMMARY

F

P

P

P₀

$$P_0 = P_A + \rho gh$$

PL

B

AP

SL & TF

Key

$$A_1 V_1 = A_2 V_2$$

✓

5 years of discipline.

50 years of freedom.

That's the formula.



Join Our Official Telegram & Whatsapp Channel



JKBOSE Wallah
Class 9th & 10th



JKBOSE Wallah



Scan the QR to Join

Thank
You

